Deepak Mallampati

APPLIED AI / FORWARD DEPLOYED ENGINEER

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US-based, F-1 OPT (open to sponsorship) - NYC-available

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years shipping production agentic LLM software directly into live, regulated operations. Designs conductor–subagent orchestration that runs autonomously (42% lower token cost at sustained accuracy), makes messy enterprise data agent-readable through large-scale RAG, and hardens it for production (60% fewer failed runs, 100k+ requests/day). Comfortable embedding with operators to translate ambiguous business needs into working software fast — the core of a forward-deployed mandate.

CORE COMPETENCIES

Agentic orchestration (conductor–subagent) • Forward-deployed / embedded delivery • RAG & data-readiness • LLM evaluation & observability • Production reliability • APIs & microservices • MLOps

Generative AI & LLMs: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Multi-Agent Systems | Prompt Engineering | Fine-tuning (LoRA/QLoRA/PEFT) | HF Transformers

Retrieval & Data: FAISS, Pinecone, Weaviate | Cross-Encoder Re-ranking | Semantic/Hybrid Search | PostgreSQL, MongoDB, Redis | Spark, Databricks, Kafka | Data Lineage & Governance

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, W&B | Monitoring & Observability

Engineering: Python, TypeScript/JavaScript, Java, SQL | FastAPI, React | RAGAS, BERTScore eval | Git, Jira

Security & Compliance: HIPAA, SOC 2, PCI DSS | PII Masking/Redaction | Prompt-Injection Mitigation | OAuth2/JWT/RBAC | Audit Logging

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 – Present | USA*

AI Engineer

- Architected production-grade **agentic LLM systems** on a conductor–subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously — reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare corpora, combining dense retrieval with cross-encoder re-ranking to lift relevance and materially reduce hallucination — turning unstructured documents into agent-readable knowledge.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to catch regressions before release — a continuous feedback loop over deployed workflows.
- Implemented **privacy-preserving data workflows** (k-anonymity, l-diversity, differential privacy) enabling compliant analytics with zero PII exposure; optimized inference cost via prompt compression, semantic caching, and model routing.

Vanguard *Jan 2024 – Jan 2025 | USA*

AI Engineer Intern

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ agents**, integrating with production data pipelines and observability tooling.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams — embedded, cross-functional delivery.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Optimized prompts and evaluated GPT-4o, Claude Sonnet, and Gemini, improving task success by **27%**; engineered content controls reducing unsafe outputs by 42%.

Kent State University *Oct 2023 – Jan 2024 | USA*

ML & Gen AI Research Assistant

- Built a privacy-preserving ML pipeline on clinical data, cutting re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an H100 with a curated instruction dataset for controllable long-form generation.

Emerson *Jun 2022 – Apr 2023 | India*

ML Trainee

- Built supervised regression/classification models for equipment-failure prediction and anomaly detection; engineered end-to-end preprocessing pipelines.
- Fine-tuned **BERT/RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER.

PROJECTS

career-ops

An autonomous, AI-driven pipeline that scans listings, scores fit, and ships tailored applications overnight without supervision — a working example of end-to-end agentic delivery from data to outcome.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites, turning a multi-step manual chore into a single inline action.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy/accuracy trade-off.

EDUCATION

M.S. in Computer Science, *Kent State University, OH* *2025 | GPA 3.70/4.0*